

Japan Autos & Auto Parts

Japan Autos & Auto Parts: Japanese automakers' interest in humanoid robots appears to be re-emerging



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Yesterday, Mitsubishi Motors announced that it had entered into an MOU with a Japanese startup to jointly develop and mass-produce humanoid robots. As is well known, Japanese automakers have a history of involvement in robotics. As the global automotive market becomes increasingly challenging, will Japanese automakers step up their efforts in robotics as an alternative growth engine?

Mitsubishi eyes humanoid robot development and mass production: Mitsubishi Motors (MMC, not covered) has signed an MOU ([link](#)) with Highlanders, a University of Tokyo spin-off specializing in humanoid robotics, to jointly develop and mass-produce humanoid robots based on Highlanders' humanoid robot platform, 'N'. The partnership includes plans to deploy robots at MMC's manufacturing facilities while leveraging the automaker's expertise in mass-production and quality assurance. Production is expected at MMC's Kyoto plant, utilizing idle factory space, with an initial target capacity of 1,000 units per month as early as 2027. MMC, which has already invested in Highlanders, also plans to increase its stake to support commercialization. As investor interest in humanoid robotics continues to grow, we view the announcement positively for the sector as it signals that Japanese automakers are taking steps toward cultivating a new mid- to long-term growth driver beyond its automotive business.

Japanese automakers have a history of involvement in robotics: Japanese automakers have a long, albeit uneven, history of involvement in robotics. Honda is widely regarded as the industry's pioneer, having begun bipedal humanoid robot research in 1986 and unveiling ASIMO in 2000, one of the world's best-known humanoids. Toyota followed with its Partner Robot program in the 2000s, targeting assistance, healthcare, and human-machine interaction applications. While these initiatives demonstrated how automotive expertise in sensing, motion control, and mechatronics could be applied beyond vehicles, robotics remained primarily a research activity rather than a commercial business.

Global humanoid robot market to reach USD 729 bn by 2025: We forecast global humanoid robot shipment to reach 49 mn units by 2050 (2025-2050 CAGR of ~37%), with the market expanding to ~USD 729 bn (~32% CAGR), making humanoid robotics an attractive growth opportunity for automakers. The sector is particularly relevant to the automotive industry given the overlap in core technologies, including actuators (motors, gear reducers, etc.), sensors, batteries, control units, and AI-enabled software, areas in which automakers and suppliers have built expertise through vehicle development. Amid intensifying competition, labor shortages, and rapid AI advances, Japanese automakers' interest in humanoid robotics appears to be re-emerging. Toyota also highlighted robotics initiatives in factories, including parts logistics and picking processes, at its latest earnings briefing. The company further indicated that it ultimately extends robotics applications beyond manufacturing into healthcare and retail. The market will likely watch for more concrete commercialization efforts.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	9 Jul 2026	Price	TTM	Reported EPS				Reported P/E (x)		
			Closing	Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
7203.JP (Toyota)	O	JPY	2,824.00	4,200.00	(31.9)%	JPY	295.25	308.93	367.29	9.6	9.1	7.7
7269.JP (Suzuki)	O	JPY	2,027.50	2,550.00	(23.3)%	JPY	227.69	234.71	255.99	8.9	8.6	7.9
7267.JP (Honda)	M	JPY	1,502.00	1,300.00	(44.9)%	JPY	(106.06)	161.84	147.24	(14.2)	9.3	10.2
7201.JP (Nissan)	U	JPY	304.50	350.00	(45.6)%	JPY	(152.58)	4.78	53.06	(2.0)	63.6	5.7
7261.JP (Mazda)	U	JPY	1,106.00	1,000.00	(15.0)%	JPY	55.64	116.61	142.17	19.9	9.5	7.8
7270.JP (Subaru)	U	JPY	2,488.50	2,350.00	(47.4)%	JPY	125.50	222.96	249.75	19.8	11.2	10.0
8015.JP (Toyota Tsusho)	O	JPY	6,110.00	8,150.00	42.5%	JPY	350.95	451.35	489.20	17.4	13.5	12.5
6902.JP (Denso)	M	JPY	1,928.00	2,050.00	(47.2)%	JPY	162.96	169.90	190.05	11.8	11.3	10.1
7259.JP (Aisin)	M	JPY	2,289.50	2,450.00	(23.8)%	JPY	232.64	248.13	236.66	9.8	9.2	9.7
JPL			2,629.48									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate **Toyota** as **Outperform** with a price target of **JPY 4,200.00**.

We rate **Suzuki** as **Outperform** with a price target of **JPY 2,550.00**.

We rate **Honda** as **Market-Perform** with a price target of **JPY 1,300.00**.

We rate **Nissan** as **Underperform** with a price target of **JPY 350.00**.

We rate **Mazda** as **Underperform** with a price target of **JPY 1,000.00**.

We rate **Subaru** as **Underperform** with a price target of **JPY 2,350.00**.

We rate **Toyota Tsusho** as **Outperform** with a price target of **JPY 8,150.00**.

We rate **Denso** as **Market-Perform** with a price target of **JPY 2,050.00**.

We rate **Aisin** as **Market-Perform** with a price target of **JPY 2,450.00**.

DETAILS

EXHIBIT 1: **Highlanders' humanoid robot 'N'**



Source: Company webstie

EXHIBIT 2: **Honda's history of robotics development**



Source: Company website

EXHIBIT 3: **Toyota's 'Mobility Robot'**



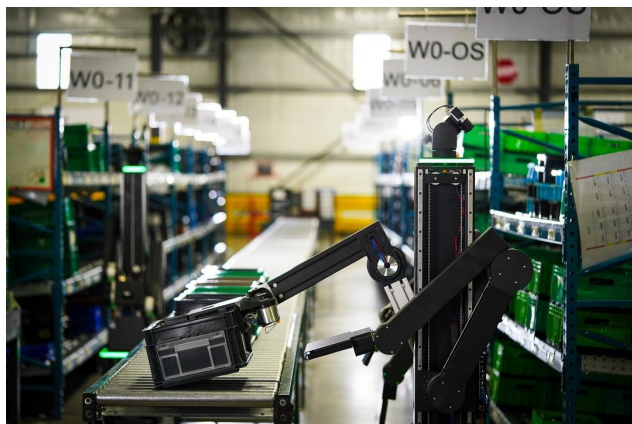
Source: Company website

EXHIBIT 4: **Toyota's 'Violin-Playing Robot'**



Source: Company website

EXHIBIT 5: **Toyota Research Institute's robots in a factory**



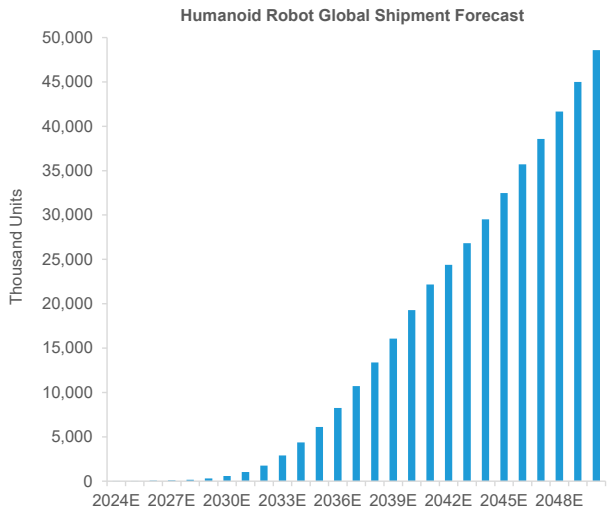
Source: Company website

EXHIBIT 6: **Toyota Research Institute's humanoid robot for mobile manipulation**



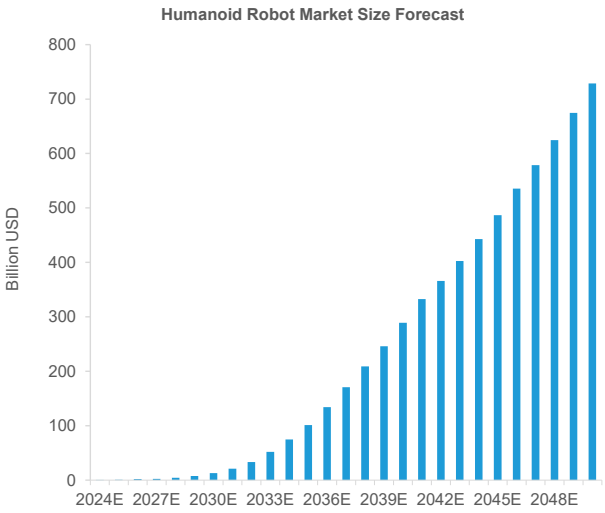
Source: Company website

EXHIBIT 7: **We forecast global humanoid robot shipment to reach 49 mn units by 2050**



Source: Bernstein's Asia Emerging Robotics team, Bernstein analysis and estimates

EXHIBIT 8: **We forecast global humanoid market size to reach USD 729 bn by 2050**



Source: Bernstein's Asia Emerging Robotics team, Bernstein analysis and estimates

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